



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO PSA FUEL POINT

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

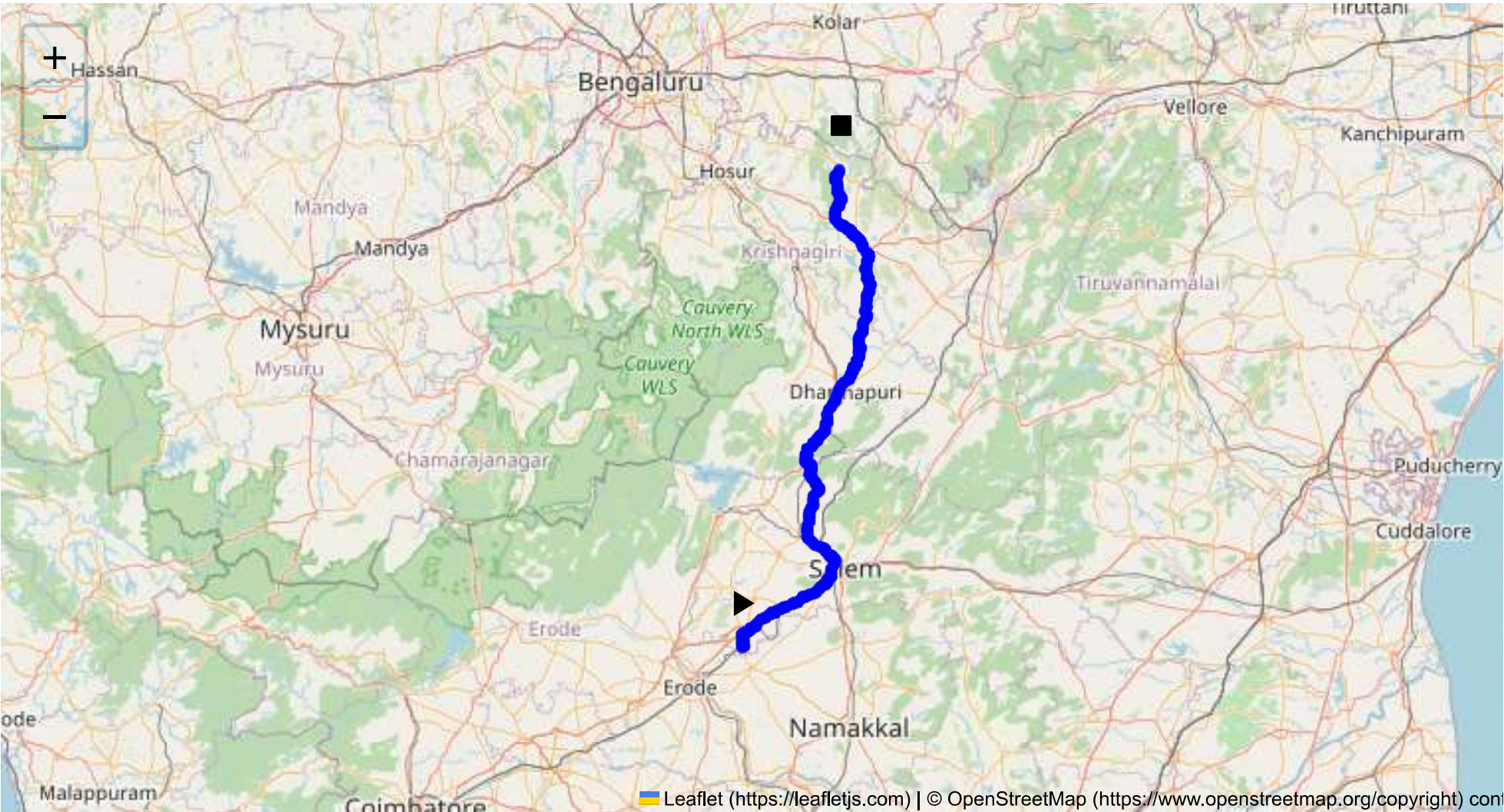
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 181.96 km
Estimated Duration: 3.3 hours
Adjusted Duration (Heavy Vehicle): 4.1 hours
Start: (11.4381, 77.8734)
End: (12.735676, 78.142759)



Welcome to the Journey Risk Management Study

Route Safety Analysis from CVQF+23W, Sangagiri, Tamil Nadu to P4PV+74, Kottakrishnapalli, Tamil Nadu

1. Overview of the Route Map

The route commences from Sangagiri and travels approximately 182 kilometers to Kottakrishnapalli, traversing through denser urban areas, rural landscapes, and hilly terrains. Major roads include National Highways and State Highways, which often connect through arterial roads in smaller towns and cities along the way.

2. Typical Weather Conditions

The route traverses a region known for its tropical climate, with hot summers, a monsoon season typically from June to September, and mild winters. During monsoon, heavy rains can cause flooding and landslides in hilly areas, potentially obstructing or impeding travel. It's essential to monitor weather forecasts, especially during monsoon seasons.

3. Traffic Patterns

Traffic is observed to be most congested during morning (8 AM - 10 AM) and evening (5 PM - 8 PM) rush hours, particularly near urban centers and at intersections with major highways. Weekends may see increased traffic due to local events or markets. Adequate planning can help navigate through these periods more efficiently.

4. Assessment of Road Quality and Infrastructure

National highways tend to have good road conditions and are well-maintained. However, areas connecting smaller villages may have narrow roads with poor surface conditions, leading to slower traffic and increased risk of incidents. Caution is advised in these segments.

5. Suggestions for Alternative Routes

In emergencies, such as road blockages or accidents, assess using route planning apps for real-time updates. Minor roads might cut through villages and should be used if confirmed safe and passable. It is advisable for drivers to familiarize themselves with two or three secondary routes as backup.

6. Summary of Local Regulations

Tamil Nadu follows strict regulations for transporting hazardous materials, requiring proper labeling, necessary permits, and adherence to prescribed transport routes. It is essential to ensure compliance with these to avoid legal or safety issues.

7. Historical Incidents

Historically, incidents involving heavy vehicles in this region have often been associated with overloading and vehicle malfunctioning. Rail crossing points are also prone to accidents. Regular vehicle checks and license renewals are necessary preventative measures.

8. Environmental Considerations and Sensitive Areas

The route passes close to farmland and water bodies. Avoid spillages and ensure no unauthorized discharge into the environment. Special care should be taken near wildlife sanctuaries or protected environments to minimize disruption.

9. Communication Coverage

The majority of the route has good mobile network coverage with leading providers. However, certain rural patches might experience connectivity issues. Drivers should be equipped with other communication means, such as radio, especially through known dead zones for coordinated communication.

10. Emergency Response Times

Response times can vary considerably across the route. Urban centers typically receive faster emergency services in under 30 minutes, whereas remote areas might take significantly longer. Keep contact numbers for local police, hospital services, and towing companies.

11. Overall Summary of Risk Assessment

This route, spanning both urban and rural terrains, presents moderate risks primarily due to weather conditions and varying road qualities. Tactical planning around peak hours, efficient emergency protocols, and facilities for route deviation are essential strategies drivers must adhere to. Mitigating the risk of transporting hazardous materials remains a priority, mandating compliance with transport regulations and prior preparation for unforeseen circumstances.

Overall, while the route is manageable with proper planning, drivers must remain alert to fluctuating road conditions and regulations, maintaining a proactive approach towards unexpected events.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.46400, 77.87323	30 KM/Hr
5	Turn	Medium	11.46765, 77.87037	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	Medium	11.47986, 77.88397	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.53512, 77.95717	30 KM/Hr
11	Turn	Medium	11.53606, 77.95869	30 KM/Hr
12	Turn	Medium	11.62403, 78.10893	30 KM/Hr
13	Turn	Medium	11.62932, 78.11659	30 KM/Hr
14	Turn	High	11.64183, 78.11810	15 KM/Hr
15	Turn	Medium	11.64338, 78.11998	30 KM/Hr
16	Turn	Medium	11.67903, 78.12796	30 KM/Hr
17	Turn	Medium	11.69532, 78.10764	30 KM/Hr
18	Turn	Medium	11.73968, 78.05098	30 KM/Hr
19	Turn	Medium	11.86790, 78.08347	30 KM/Hr
20	Turn	Medium	11.91357, 78.05868	30 KM/Hr
21	Turn	Medium	11.92556, 78.06499	30 KM/Hr
22	Turn	Medium	11.93296, 78.06432	30 KM/Hr
23	Turn	Medium	11.93535, 78.06150	30 KM/Hr
24	Turn	Medium	11.95531, 78.04698	30 KM/Hr
25	Turn	High	11.95803, 78.04964	15 KM/Hr
26	Turn	Medium	11.96068, 78.04960	30 KM/Hr
27	Turn	High	11.96370, 78.05142	15 KM/Hr
28	Turn	High	11.96542, 78.04938	15 KM/Hr
29	Turn	Medium	11.96650, 78.04927	30 KM/Hr
30	Turn	Medium	11.96890, 78.05109	30 KM/Hr
31	Turn	Medium	12.22422, 78.19127	30 KM/Hr
32	Turn	Medium	12.24685, 78.19561	30 KM/Hr
33	Turn	Medium	12.25012, 78.19946	30 KM/Hr
34	Turn	Medium	12.25280, 78.20041	30 KM/Hr
35	Turn	Medium	12.29065, 78.19892	30 KM/Hr
36	Turn	Medium	12.32012, 78.20734	30 KM/Hr
37	Turn	Medium	12.40099, 78.21805	30 KM/Hr
38	Turn	Medium	12.45136, 78.21799	30 KM/Hr
39	Turn	Medium	12.50485, 78.22336	30 KM/Hr
40	Turn	Medium	12.59710, 78.13638	30 KM/Hr
41	Turn	High	12.60040, 78.12942	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
42	Turn	High	12.62082, 78.13182	15 KM/Hr
43	Turn	Medium	12.62152, 78.13113	30 KM/Hr
44	Turn	High	12.62315, 78.13063	15 KM/Hr
45	Turn	High	12.62382, 78.12923	15 KM/Hr
46	Turn	Medium	12.63690, 78.13736	30 KM/Hr
47	Turn	Medium	12.66185, 78.14415	30 KM/Hr
48	Turn	Medium	12.66560, 78.14270	30 KM/Hr
49	Turn	Medium	12.68017, 78.13450	30 KM/Hr
50	Turn	High	12.70625, 78.13518	15 KM/Hr
51	Turn	High	12.70934, 78.13064	15 KM/Hr
52	Turn	High	12.71821, 78.12989	15 KM/Hr
53	Turn	Medium	12.71981, 78.13322	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
6	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
8	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
9	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
10	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
17	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
18	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
19	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
21	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
22	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
23	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
24	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
25	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
26	hospital	Global Medical Centre Jagirammapalayam	11.6894409, 78.1108847	30 km/h	Medium
27	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
31	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
32	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
35	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
36	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
37	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
39	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
44	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium
45	hospital	St. Mary's Hospital	12.5128555, 78.2186887	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
46	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
47	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
48	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium
49	clinic	Shree Ramachandra ENT Centre	12.5290096, 78.2037359	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
11	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
13	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
14	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
15	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
16	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
20	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
28	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
29	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
30	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
33	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
34	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
38	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium